

Research on Parameter Adaptive Control Strategy of Virtual Synchronous Generator

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Abstract—The virtual synchronous generator control introduces the synchronous generator rotor motion equation into the control of the grid-connected inverter to realize the friendly grid connection of the distributed power source. However, a part of the dynamic adjustment performance of the grid-connected inverter is lost. Improper selection of the moment of inertia and damping coefficient of the virtual synchronous generator may cause the "rocking" oscillation phenomenon, which is similar to the synchronous generator, appearing in the grid-connected inverter, and even cause the grid-connected inverter to be off-grid. Therefore, the virtual synchronous generator parameter adaptive control is proposed for the moment of inertia and damping coefficient. Firstly, the mathematical model of the virtual synchronous generator is established, and the influences of moment of inertia and damping coefficient on the performances of virtual synchronous generator is analyzed. Furthermore, the adaptive control of moment of inertia and damping coefficient is introduced into the virtual synchronous generator control. Finally, the effectiveness of the proposed control strategy is verified by MATLAB/Simulink simulation. The results show that the parameter adaptive virtual synchronous generator control improves the system response speed and reduces effectively the virtual synchronous generator frequency and power overshoot, compared with the fixed-parameter virtual synchronous generator control.

Keywords—Virtual synchronous generator, Parameter adaptive control, Moment of inertia, Damping coefficient, Rocking oscillation

I. INTRODUCTION

With the increasingly prominent fossil energy crisis and environmental problems, renewable energy is growing rapidly around the world, more and more photovoltaic power generation unit and wind turbines are connected to the grid, the power system is changing from traditional centralized power generation to distributed power generation [1-2], the form of distributed power output electrical energy is different from grid,

therefore, most of the distributed power source grid-connected need to be connected to the grid through power electronic converters. Power electronic converters have the characteristics of flexible control and short transient time [3], however, due to the lack of rotational inertia and damping effects of traditional synchronous generators [4], its ability to suppress interference and fluctuations is weak. For this reason, in recent years, scholars at home and abroad have proposed a virtual synchronous generator (VSG) control strategy for grid-connected inverters, introducing synchronous generator rotor motion equation into the under control of grid-connected inverter, making the grid-connected inverter have the moment of inertia and damping of the synchronous generator, thereby enhances the ability of grid-connected inverters to suppress output power and frequency fluctuations, enhancing system stability.

The VSG control makes the grid-connected inverter lose part of the dynamic regulation performance, so the control parameters of the VSG have a crucial impact on the performance of the grid-connected inverter. Reference [5] proposes an adaptive inertial control strategy to achieve a balance between power regulation and frequency regulation according to different working conditions, and gives the design principle of parameters to determine the range of adaptation. Reference [6] proposes an adaptive control strategy to keep the system at the best damping ratio during operation to optimize overshoot and response time and other indicators, thereby suppressing frequency and power oscillations and improving the dynamic performance of the VSG. Reference [7] proposed an adaptive virtual inertial control method using fuzzy logic to ensure the frequency stability required for the system to operate under high penetration conditions. However, this control method requires a large amount of calculation and has difficulties in practical application.

In this paper, the mathematical model of VSG is firstly established to analyze the influence of moment of inertia and damping coefficient on VSG performance. According to the

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moment of inertia and damping coefficient, proposing an adaptive control strategy for VSG parameters. When the VSG voltage angular frequency acceleration deviates from the rated value, increase the moment of inertia and damping coefficient. When the voltage angular frequency decelerates close to the rated value, reducing the moment of inertia and increase the damping coefficient. Finally, the effectiveness of the proposed control strategy is verified by MATLAB/Simulink simulation and RT-lab hardware-in-the-loop experiments. The results show that the proposed adaptive control strategy reduces the VSG output power compared with the fixed-parameter VSG control. The results show that, compared with the fixed parameter VSG control and amount of frequency overshoot, the response speed of the system is improved, and the small-amplitude "rocking" oscillation of the output power is avoided.

II. VSG ACTIVE POWER LOOP TRANSFER FUNCTION

Fig. 1 is a diagram of the grid-connected inverter topology and VSG control structure. The electrical part includes a DC voltage source, a of DC Voltage Stabilizer Capacitor, a three-phase inverter bridge, an LC filter and an equivalent grid; The control part includes VSG control, Q-V droop control, virtual impedance control and voltage and current double-loop control. The Q-V droop control enables the grid-connected inverter to have voltage support capability.

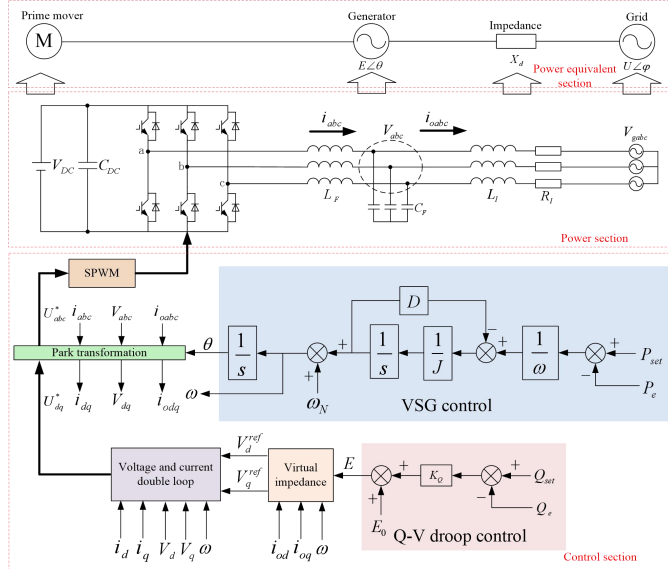


Fig. 1. Grid-connected inverter topology and VSG control structure diagram

Since the reactive power control part has nothing to do with the research content of this paper, it will not be repeated. The virtual impedance control and voltage and current double-loop control reduce the output voltage and current harmonics of the grid-connected inverter, and reduce the influence of the inverter parameter perturbation on the output performance of the converter [8]. The response speed of virtual impedance control and voltage-current double-loop control is faster. Therefore, it can be ignored in the mathematical model of VSG [9].

The active power control part introduces VSG control, and adopts the classical second-order model of synchronous generator, as shown in (1).

$$\begin{cases} J \frac{d\omega}{dt} = \frac{P_{set} - P_e}{\omega} - D(\omega - \omega_N) \\ \frac{d\theta}{dt} = \omega \end{cases} \quad (1)$$

Where J is the VSG moment of inertia, D is the VSG damping coefficient, P_{set} is the VSG output power set value, P_e is the VSG output power actual value, ω_N is the VSG voltage rated angular frequency, ω is the VSG actual voltage angular frequency, θ is the VSG power angle.

(1) can be further rewritten as:

$$\begin{cases} J \frac{\Delta\omega}{\Delta t} = \frac{P_{set} - P_e}{\omega} - D(\omega - \omega_N) \\ \frac{d\theta}{dt} = \omega \end{cases} \quad (2)$$

Where $\Delta\omega = \omega - \omega_N$, after Laplace transform of (2), (3) is obtained:

$$\begin{cases} \omega = \frac{P_{set} - P_e}{\omega} \frac{1}{Js + D} + \omega_N \\ \theta s = \omega \end{cases} \quad (3)$$

The first equation in (3) can be written as:

$$\omega_\delta = \omega - \omega_g = \frac{P_{set} - P_e}{\omega} \frac{1}{Js + D} + \omega_N - \omega_g \quad (4)$$

Where ω_g is the grid voltage angular frequency, ω_δ is the difference between the VSG voltage angular frequency and the grid voltage angular frequency.

Known:

$$\omega_\delta = \frac{d\delta}{dt} \quad (5)$$

Where δ is the phase angle between the VSG voltage and the grid voltage.

After performing Laplace transform on (5), got:

$$\omega_\delta = \delta s \quad (6)$$

The active power calculation is shown in (7):

$$P_e = \frac{EU}{X_d} \sin(\theta - \varphi) = \frac{EU}{X_d} \sin \delta \approx \frac{EU}{X_d} \delta \quad (7)$$

Among them, E 、 θ are the effective value and phase angle of the VSG output voltage, respectively, and U 、 φ are the effective value and phase angle of the grid voltage, respectively. X_d is the VSG synchronous reactance.

According to (4), (6) and (7), the active power control structure under the VSG grid-connected operation can be obtained as shown in Fig. 2.

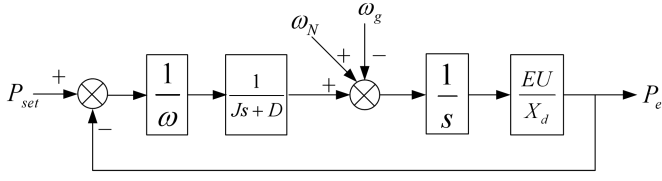


Fig. 2. Active power control structure under VSG on-grid operation

Through Fig. 2, the open-loop transfer function (8) and closed-loop transfer function (9) corresponding to the active power control ring during VSG grid connected operation can be obtained.

$$G_{p_o}(s) = \frac{\frac{EU}{J\omega X_d}}{s\left(s + \frac{D}{J}\right)} \quad (8)$$

$$G_{p_c}(s) = \frac{P_e}{P_{set}} = \frac{\frac{EU}{J\omega X_d}}{s^2 + \frac{D}{J}s + \frac{EU}{J\omega X_d}} \quad (9)$$

III. INFLUENCE OF MOMENT OF INERTIA J AND DAMPING COEFFICIENT D ON VSG CHARACTERISTICS

According to (9), we can know that the system output active dynamic response under different moment of inertia and damping coefficient is shown in Fig. 3 and Fig. 4. According to Fig. 3 and Fig. 4, when D is constant, the large J is, the larger the oscillation frequency of the system output active power response process is, and the slower the system response speed is. When J is constant, the larger D is, the faster the system output active power oscillation decays

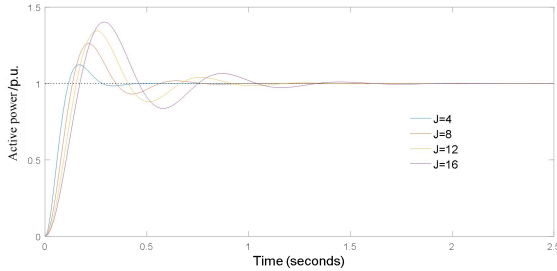


Fig. 3. System output active dynamic response under different moments of inertia J ($D = 100$)

The VSG control strategy makes the grid-connected inverter have the active power-frequency droop external characteristics similar to the synchronous generator [4], and the sudden change of VSG active power will cause the VSG angular frequency to change. The first formula in (2) can be further rewritten after Laplace transform as:

$$G_{\omega-P}(s) = \frac{\Delta\omega}{\Delta P} = \frac{1}{\omega} \frac{1}{Js + D} \quad (10)$$

Where $\Delta P = P_{set} - P_e$. We can know from (10), the transfer function of VSG active power deviation and angular frequency deviation is first-order delay link. Fig. 5 and Fig. 6 is the dynamic response of VSG angular frequency deviation under different moment of inertia and damping coefficient. According to Fig. 5 and Fig. 6, when D is constant, the smaller J is, the faster the dynamic response speed of VSG angular frequency deviation is. When J , the larger the D , the smaller the VSG angular frequency deviation.

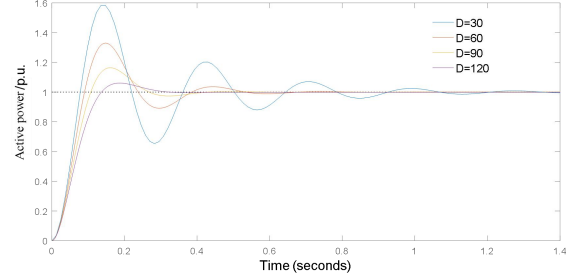


Fig. 4. System output active dynamic response under different damping coefficients D ($J = 4$)

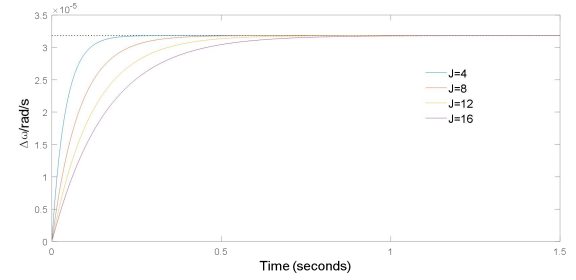


Fig. 5. Dynamic response of VSG angular frequency deviation under different moments of inertia J ($D = 100$)

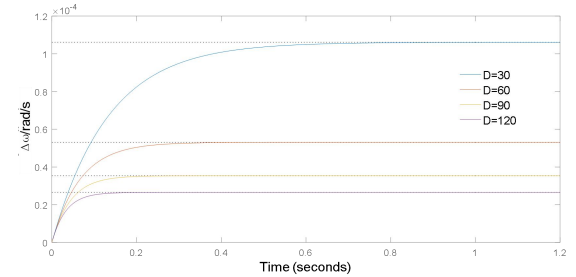


Fig. 6. Dynamic response of VSG angular frequency deviation under different damping coefficients D ($J = 4$)

Based on the above analysis, it can be seen that the larger the moment of inertia of the VSG, the larger the system amount of overshoot, the longer the adjustment time and the slower the system responding speed; The larger the damping coefficient of the VSG, the smaller the amount of system overshoot, the shorter the adjustment time, and the smaller the VSG angular frequency deviation. Therefore, when designing the adaptive control strategy of VSG moment of inertia J and damping coefficient D , we should not only avoid the system

instability caused by improper parameter design, but also make full use of the influence of control parameters on the performance of VSG to optimize the operation of VSG.

IV. VSG PARAMETER ADAPTIVE CONTROL STRATEGY

A. Analysis of Dynamic Adjustment Process of VSG

When the VSG happens a sudden load change or receives an previous layer scheduling instruction and other conditions, the system will enter another stable state from one stable state, and the power angle and voltage angular frequency of the VSG will oscillate. Taking the sudden change of output active power set value P_{set} abruptly as an example, at 5s, P_{set} changes from 0.4p.u to 0.8p.u., the difference between the VSG voltage angular frequency of the fixed control parameter and the VSG voltage rated angular frequency $\Delta\omega$ and the VSG voltage angular frequency change rate $d\omega/dt$ of curve with time is shown in Fig. 7.

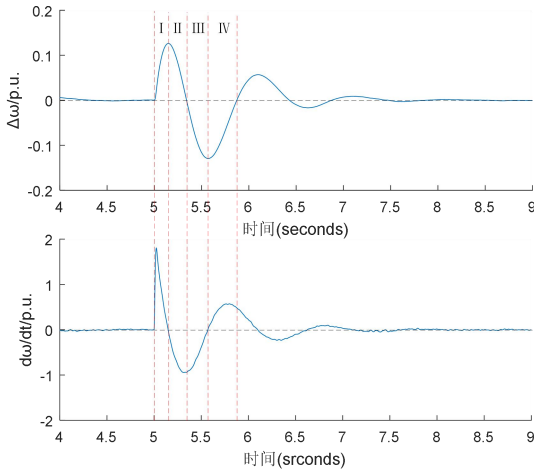


Fig. 7. The change process of $\Delta\omega$ and $d\omega/dt$

It can be seen from Fig. 7 that an oscillation period of $\Delta\omega$ and $d\omega/dt$ can be divided into 4 stages of response process: In process I, P_{set} suddenly increases, in equivalent Fig. 1, sudden increase of mechanical power output of prime mover, this causes the VSG voltage angular frequency to rise rapidly. At this time, $\Delta\omega$ gradually increases from 0 to the difference peak value, and $d\omega/dt$ suddenly increases from 0 to the peak value gradually decreases to 0, and the VSG is in the positive acceleration stage at this time; In the process II, $\Delta\omega$ gradually decreases from the peak value of the difference to 0, and $d\omega/dt$ gradually decreases from 0 to the valley value. At this time, the VSG is in the positive direction deceleration stage; In the process III, $\Delta\omega$ gradually decreased from 0 to the valley value of the difference, and $d\omega/dt$ gradually increased from the valley value to 0, at this time VSG was in the reverse acceleration stage; During the process IV, $\Delta\omega$ raised from the valley value difference gradually to 0, and $d\omega/dt$ gradually add from 0 to the peak then decreases, which is caused by the oscillation decay period becomes longer. At this time, the VSG

is in the reverse deceleration stage. The next oscillation cycle continues to repeat the 4-stage response process until the amplitude decays to 0, and the system enters the next stable state.

B. Design of Adaptive Control Strategy for VSG Parameters

The purpose of the VSG parameter adaptive control strategy is though control the VSG acceleration process and deceleration process, so that the frequency and power dynamic response speed is accelerated, the amount of overshoot is reduced and able to transition smoothly and quickly from one stable state to another stable state. In process I, VSG needs to reduce the system response speed to make peak value reduce. Therefore, the moment of inertia is appropriately increased to slow down the response speed of the system. At the same time, the damping coefficient is increased to avoid the increase of the system overshoot caused by the increase of the moment of inertia and reduce VSG angular frequency deviation. In process II, the VSG needs to improve the system response speed so that the VSG deviation of angular frequency can quickly recovery to 0. Therefore, the moment of inertia is appropriately reduced to accelerate the system response speed, at the same time, properly increase the damping coefficient to avoid excessive change rate of voltage angular frequency when VSG angular frequency deviation passes through point of zero. The system response modes of process III and process IV are the same as those of process I and process II respectively, just in the opposite direction. Therefore, the changing principles of moment of inertia and damping coefficient are the same, and will not be repeated.

The transient response process of VSG in one oscillation period is shown in Table 1.

TABLE I. RESPONSE PROCESS OF VSG DURING THE TRANSIENT

Process	$\Delta\omega$	$d\omega/dt$	State	Change in J	Change in D
I	>0	>0	Positive acceleration	Increase	Increase
II	>0	<0	Positive deceleration	Reduce	Increase
III	<0	>0	Reverse acceleration	Increase	Increase
IV	<0	<0	Reverse deceleration	Reduce	Increase

According to the analysis in Table 1, the changes of moment of inertia J and damping coefficient D are positively and negatively related to $\Delta\omega$ and $d\omega/dt$. Therefore, according to the changes of $\Delta\omega$ and $d\omega/dt$ designing VSG Parameter Adaptive Control Strategy such as (11), (12) shown.

Among them, J_0 and D_0 are the moment of inertia and damping coefficient when the VSG runs stably, respectively, and K_J and K_D are the accommodation coefficient of the moment of inertia and damping coefficient, respectively. In order to prevent frequent changes of control parameters when VSG is running stably and affect system performance, and L_J and L_D are set as the VSG parameter adaptive control trigger

thresholds. When $|\Delta\omega|$ is less than each threshold, the moment of inertia and damping coefficient are fixed values.

$$J = \begin{cases} J_0 & |\Delta\omega| < L_J \\ J_0 e^{K_J(d\omega/dt)} & \text{Process I, II} \\ J_0 e^{-K_J(d\omega/dt)} & \text{Process III, IV} \end{cases} \quad (11)$$

$$D = \begin{cases} D_0 & |\Delta\omega| < L_D \\ D_0 e^{K_D|\Delta\omega|} & \text{Process I, II, III, IV} \end{cases} \quad (12)$$

C. VSG Parameter Adaptive Control Parameter Tuning

For moment of inertia J according to the principle that the system will be stable only when the two characteristic roots of the second-order system in the control theory are in the left half complex plane, determining the minimum value of moment of inertia $J_{\min}$ and determining the maximum moment of inertia according to the principle of the maximum power output capability of the grid-connected inverter $J_{\max}$, thus, the moment of inertia can be selected flexibly J_0 . For the damping coefficient D_0 when the VSG runs stably, it is set according to the method of "optimal second-order system"[4].

According to the closed-loop transfer function (9) of the active power control loop, its characteristic equation can be obtained as:

$$s^2 + \frac{D}{J}s + \frac{EU}{J\omega X_d} = 0 \quad (13)$$

The characteristic root of the characteristic equation is:

$$s_{1,2} = \frac{-\frac{D}{J} \pm \sqrt{\frac{D^2}{J^2} - \frac{4EU}{J\omega X_d}}}{2} \quad (14)$$

According to the open-loop transfer function (8) of the active control loop, the root locus of the VSG under different control parameter changes is drawn as shown in Fig. 8:

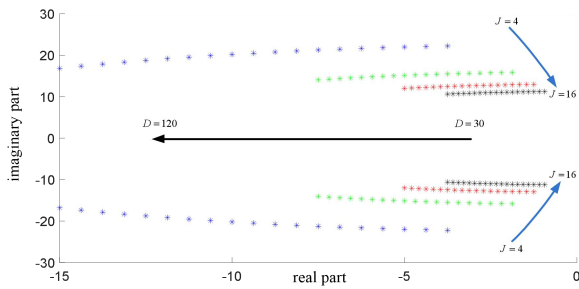


Fig. 8. Root locus under variation of different parameters ($J: 4 \sim 20, D: 30 \sim 150$)

As can be seen from Fig. 8, when J is the same, the larger the D is, the smaller the characteristic root s_1, s_2 the real part, the smaller the absolute value of the imaginary part, which indicates that the stability of the system becomes better but the response speed becomes slower; When D is the same, the larger the J is, the larger the real part of the characteristic

root s_1, s_2 is, and the smaller the absolute value of the imaginary part is, which indicates that the stability of the system becomes worse, but the response speed becomes slower. Therefore, considering all aspects of limiting factors, taking: $J = 4, D = 127.5$.

V. SIMULATION ANALYSIS

In order to verify the effectiveness of the VSG parameter adaptive control strategy proposed in this paper, a 10kW VSG model as shown in Fig. 1 was built in MATLAB/Simulink software, and the designed VSG parameter adaptive control strategy was used. See Table A1 in Appendix A for simulation parameters. when the model is set at 5 seconds, the output active power set value P_{set} is abruptly changed from 0.4p.u to 0.8p.u..

Fig. 9 is the curve over time of $\Delta\omega$ and $d\omega/dt$ of the VSG fixed parameter control strategy and the parameter-adaptive control strategy.

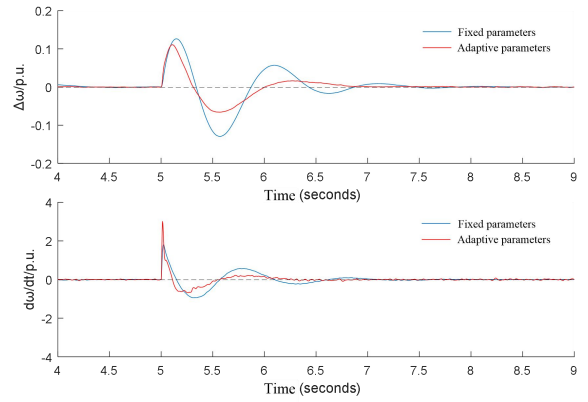


Fig. 9. The change process of $\Delta\omega$ and $d\omega/dt$

It can be seen from Fig. 9 that, compared with the VSG fixed parameter control strategy, after adopting the VSG parameter adaptive control strategy, when the VSG is in the positive acceleration state, the peak value of $\Delta\omega$ decreases. Because of $d\omega/dt$ suddenly increases, its peak value increases, making $\Delta\omega$ reach the peak value faster. In the forward deceleration state of VSG, $\Delta\omega$ decreases to 0 more quickly. The system adjustment time is reduced, and the system responds faster.

Fig. 10 shows the output active power P_e curve over time of VSG constant parameter control strategy and parameter adaptive control strategy.

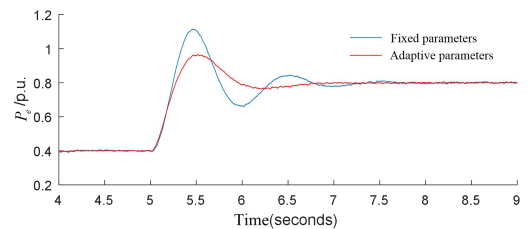


Fig.10 The change process of P_e

We can know from Fig. 10, the active power overshoot using VSG constant parameter control and parameter adaptive control are 39.36% and 21.30%, respectively, and the adjustment time is about 1.56 s and 1.46 s, respectively.

Fig. 11 gives the variation of J and D under the VSG parameter adaptive control strategy. The reason of the suddenly vertical changes of J and D in the red circled part in Fig. 11 is that $|\Delta\omega|$ does not exceed the respective set trigger thresholds.

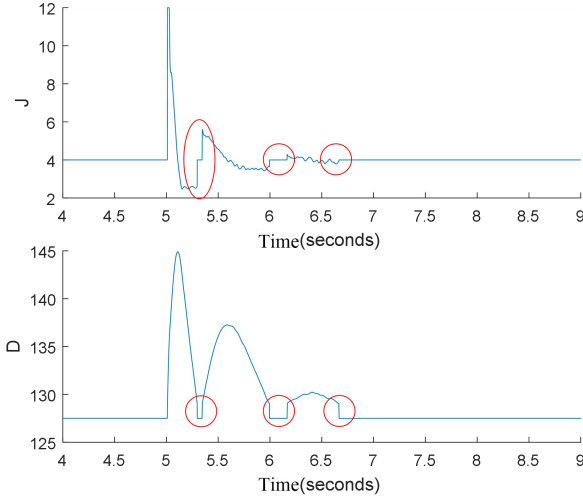


Fig.11 Variation of moments of inertia and damping coefficients

VI. CONCLUSION

When the VSG has a sudden load change or receives an upper layer scheduling command and other conditions, improper selection of the VSG moment of inertia and damping coefficient may cause the grid-connected inverter to appear "rocking" oscillation phenomenon similar to a synchronous generator. In this paper, by studying the control method of VSG, a VSG parameter adaptive control strategy is proposed for the moment of inertia and damping coefficient, and the conclusions are as follows:

1) By establishing a VSG mathematical model, the open-loop transfer function and closed-loop transfer function corresponding to the active power control ring during grid connected operation are obtained, then, getting the influence of moment of inertia J and damping coefficient D on VSG characteristics. And calculate the moment of inertia and damping coefficient when the VSG runs stably.

2) Compared with the fixed parameter VSG, the parameter adaptive VSG reduces the overshoot of the VSG output frequency and power, shortens the adjustment time of the output power, improves the system response speed, and avoids output power appearing small range "rocking" oscillations.

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APPENDIX A

TABLE A1 RESPONSE PROCESS OF VSG DURING THE TRANSIENT

Rated power S_{base}	10kVA
Rated voltage V_{base}	380V
Rated frequency f_{base}	50Hz
Rated DC voltage V_{DCbase}	700V
DC voltage V_{DC}	1.0p.u.
DC capacitance C_{DC}	0.12p.u.
Grid voltage V_{gabc}	1.0p.u.
Filter inductor L_F	0.8p.u.
Filter capacitor C_F	0.01p.u.
Line equivalent inductance L_l	0.15p.u.
Line equivalent resistance R_l	0.01p.u.
VSG voltage rated corner frequency ω_N	1.0p.u.
Voltage setting value E_0	1.0p.u.
Reactive power setting value Q_{set}	0
Q-V droop factor K_Q	0.1
Virtual impedance R_v, X_v	0.005, 0.035
Voltage loop PI control parameters K_{pv}, K_{iv}	2, 5
Current loop PI control parameters K_{pi}, K_{ii}	2, 5
PWM switching frequency f_s	10kHz